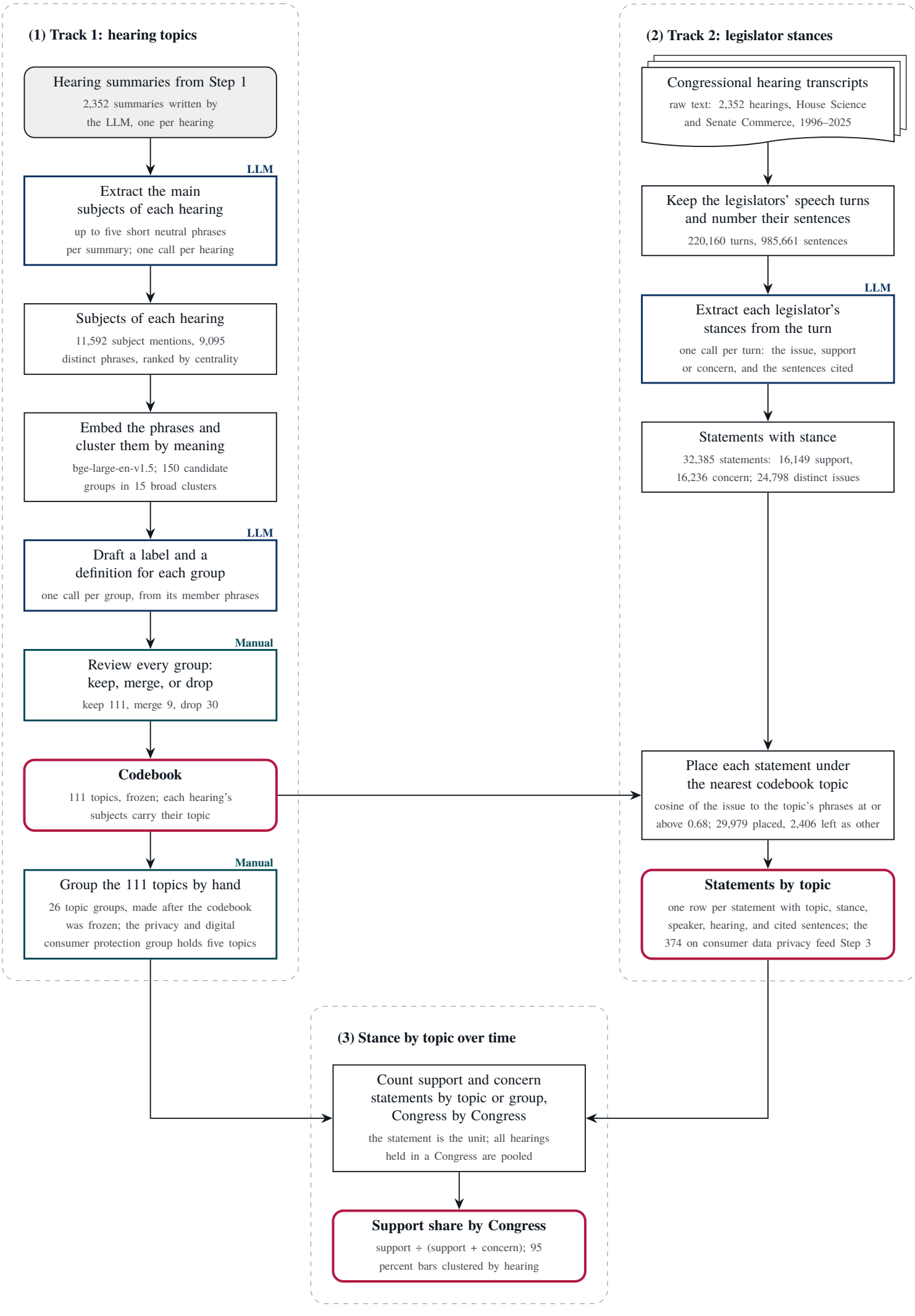

(1) Track 1: hearing topics
Hearing summaries from Step 1
2,352 summaries written by the LLM, one per hearing
LLM
Extract the main subjects of each hearing
up to five short neutral phrases per summary; one call per hearing
Subjects of each hearing
11,592 subject mentions, 9,095 distinct phrases, ranked by centrality
Embed the phrases and cluster them by meaning
bge-large-en-v1.5; 150 candidate groups in 15 broad clusters
LLM
Draft a label and a definition for each group
one call per group, from its member phrases
Manual
Review every group: keep, merge, or drop
keep 111, merge 9, drop 30
Codebook
111 topics, frozen; each hearing's subjects carry their topic
Manual
Group the 111 topics by hand
26 topic groups, made after the codebook was frozen; the privacy and digital consumer protection group holds five topics
(2) Track 2: legislator stances
Congressional hearing transcripts
raw text: 2,352 hearings, House Science and Senate Commerce, 1996–2025
Keep the legislators' speech turns and number their sentences
220,160 turns, 985,661 sentences
LLM
Extract each legislator's stances from the turn
one call per turn: the issue, support or concern, and the sentences cited
Statements with stance
32,385 statements: 16,149 support, 16,236 concern; 24,798 distinct issues
Place each statement under the nearest codebook topic
cosine of the issue to the topic's phrases at or above 0.68; 29,979 placed, 2,406 left as other
Statements by topic
one row per statement with topic, stance, speaker, hearing, and cited sentences; the 374 on consumer data privacy feed Step 3
(3) Stance by topic over time
Count support and concern statements by topic or group, Congress by Congress
the statement is the unit; all hearings held in a Congress are pooled
Support share by Congress
support ÷ (support + concern); 95 percent bars clustered by hearing
Raw text documents
Extracted phrases and statements
Text written by the LLM
Output of the step
LLM step (Qwen3.5-4B)
Manual step
Computation without a model call